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Anti-corruption campaign and economic growth in Chinese cities: The dichotomous mechanism of network-based political competition[☆]

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ABSTRACT

I study how economic growth in Chinese cities responds to the recent anti-corruption campaign, focusing on a novel mechanism of network-based political competition, whereby the removal of investigated officials creates job vacancies and triggers promotion competition within the social network of the investigated officials for the vacancies. Such a network-based competition hinges on the premise that (1) some positions are persistently occupied by certain networks, and (2) high-ranked officials help low-ranked ones to achieve promotion within the network, two facts that are well supported by the evidence. Using a difference-in-differences and an instrumental variable approach, I find that cities' GDP growth rate registers a 2.5-percentage-point increase following the investigation of the city leader's connected officials. City leaders create a healthier business environment, attract investments, and increase government spending, while some long-term issues, including innovation, education, and environmental protection, are compromised, indicating the dichotomous nature of political competition.

1. Introduction

The governance of cities is the crux of economic growth and is an outcome of city leaders' performance and policy choices, which, in turn, are shaped by their incentives. In democratic regimes, city leaders are appointed via elections and respond to voters' demands; in autocratic regimes such as China, city leaders are appointed by their superiors and react to any factors that affect their promotion prospects. In this paper, I study how Chinese city leaders respond to promotion incentives yielded by anti-corruption shocks that are transmitted in their social networks (or political factions), which are core determinants of the implementation of city governance and the outcome of economic growth. While the main body of papers on (anti)corruption and economic growth use cross-country data (Lambsdorff, 2006), I use a city data set and a new data set of the comprehensive information on social networks of city leaders, to illustrate the role of a novel politico-economic mechanism – network-based political competition for promotion – in shaping the effects of (anti)corruption on economic growth, at the city level. I find that social connections, interacting with the anti-corruption campaign, can have both a bright side (boosting economic growth) and a dark side (overlooking long-term issues

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such as innovation, education, and environmental protection), pointing to the dichotomous nature of politicians' social networks, political competition, and anti-corruption.

To be more specific, I examine how having social connections (or factional linkages) with officials who have been investigated for corruption impacts a city leader's economic performance (measured by economic growth) and policy outcomes. I do not focus on the effects of investigations of city leaders on the economic growth of that city, because such cases are too scarce to provide sufficient statistical power. The investigation and resulting dismissal of officials during the recent anti-corruption crackdown in China has created job vacancies. As a result, the remaining officials may work harder to compete for these vacancies. Such competition occurs within the social network of the removed officials since the job turnover cost within a social network is smaller for two reasons: (1) some positions are persistently occupied by certain networks, a phenomenon called "network persistence" in this paper (Reiter, 2004); and (2) higher-ranked officials may help lower-ranked ones in the same network to achieve promotion, a phenomenon called "intra-network support" in this paper (Francois et al., 2016). These two features are well supported by anecdotal facts and empirical evidence,[1] and are two essential aspects of Chinese politics.

(Anti-)corruption is an important factor for economic growth, but its effects are still under heated academic debate. To create a healthy political environment, many countries have implemented stringent anti-corruption policies. China, as a case in point, started a nationwide anti-corruption campaign in 2012. While there is a huge body of literature on the effects of corruption and anti-corruption on economic growth, the findings are mixed and, they mostly use country-level data. Regional-level studies, such as city-level within a country, are scarce. Even fewer studies analyze and provide evidence of the micro-level politico-economic mechanisms of the effects of (anti)corruption on growth. The only exception is Qu et al. (2018), who find that anti-corruption campaign has negative impacts on economic growth at the provincial level. This paper tends to fill this gap, by examining the effects of China's 2012 anti-corruption campaign on economic growth at the city level. In particular, I argue that local leaders' social networks are playing a crucial role in shaping the outcome of regional economic growth under anti-corruption campaigns.[2]

An important institutional feature is that it actually takes a long time (more than half a year) for the job openings to be filled by new people, thus leaving the relevant politicians enough time to respond. Moreover, the creation of new job openings may have persistent effects since the creation of one new job opening leads to the creation of another (an opening, say job A, is created, and, thus, after the person in job B fills the opening for A, the opening for B is now created). This results in a "chain reaction". Another institutional feature is that economic performance, even in the short run, is highly valued when local leaders are evaluated for promotion by their superiors (Li and Zhou, 2005), and such fact is supported by empirical results. Such competition is especially more intriguing in the context of politicians' social networks,[3] strategically interact with one another. The incentive to compete is more salient for officials within a social network, since the job turnover cost for the participants in the promotion competition is smaller because, first, some positions are persistently held by officials from the same network (Reiter, 2004), and, second, because the officials at higher levels in the bureaucratic pyramid can help those at lower levels within the same network to achieve promotion (Francois et al., 2016). In this paper, I analyze whether the recent anti-corruption crackdown in China on social peers matters for economic and policy outcomes via intra-network competition, the main channel in the empirical analysis, using data from the first five-year period after 2012.

I focus on several types of social ties that have been featured in earlier work in the social sciences: hometown (i.e., prefecture city) ties, college ties, and past employment relationships.[4] To alleviate endogeneity, I use the connections established prior to being in office as the measure for social connections, since they are predetermined.

I begin with the preferred specification that includes fixed effects for cities, years, and shared hometowns, colleges, and workplaces, based on a city-year level longitudinal data set, and, thus, the specification is a generalized difference-in-differences strategy. I argue that the inclusion of these fixed effects is important in distinguishing the role of shared backgrounds from unobserved differences in officials with shared attributes. The baseline results show that having a college tie or a workplace tie to an investigated corrupt official increases the per capita GDP growth rate by 2.5 percentage points, about 28% of the sample mean. Even controlling for city leader fixed effects yields qualitatively similar results. I use an event-study design to rule out the possibility that the baseline results are driven by unobserved confounding pre-treatment trends. In addition, the results still hold when I use the targets of the Central Inspection Team (CIT) as the instrumental variable. The targets may serve as a valid instrument since they might be exogenous, conditional on controls and fixed effects, especially the year fixed effects that control for all decisions by the central government and the Politburo, and the social group fixed effects that control for all time-invariant factors that may attract the attention from the CIT.[5] The sign of the partial effects, together with additional results, favors the hypothesis of intra-network

[1] I will discuss them in detail in Section 3.

[2] The investigation of corrupt officials in the campaign can be deemed a shock due to its secrecy beforehand and abruptness of implementation, which is exogenous to economic growth and policy outcomes. Officials who have engaged in corruption before might feel anxious all the time during the anti-corruption campaign, because they are uncertain of whether and when they will be investigated and penalized (http://news.sina.com.cn/o/2014-07-15/073030520696.shtml). Hence, its causal effects can be well identified, which distinguishes it from other causes of job turnover (such as retirement). It also provides a unique setting to evaluate how politicians respond to promotion incentives yielded by such shocks. As a result, this paper estimates the causal impacts of the anti-corruption crackdown on economic growth in China—the world's second most populous country and second largest economy—through the channel of intra-network competition.

[3] Like "Shaanxi Gang", those who were born or have worked in Shaanxi province, and "Tsinghua Gang", those who graduated from Tsinghua University.

[4] Some papers, including Jiang and Zhang (2020), focus on the linkages between provincial Party Secretaries and the city leaders that they promoted. However, I do not use such linkages in my paper. The first reason is that the percentage of provincial Party Secretaries who have been investigated for corruption is very small. The second reason is that such linkages do not reflect the politicians' social networks and, thus, are not consistent with the theme of this paper.

[5] The instrument may not be valid without such controls and fixed effects.

competition. Finally, the main results are not driven by the worse economic performance of investigated city officials compared to the connected officials.

The results herein are seemingly inconsistent with Qu et al. (2018), who find that the anti-corruption campaign negatively affects the Chinese economy. However, they are estimating the overall effects of the campaign on the economy, while I am only documenting a specific channel or an unintended consequence—campaign-induced political competition. While the effects through this specific channel are positive, it is still possible that the campaign affects the economy via other channels, thus leading to overall negative impacts.

The following question is: how do officials achieve their goals of enhancing economic growth? The answer is threefold: by creating a healthier business environment, actively attracting investment, and increasing public spending. The first policy instrument is effective for encouraging business creation and firm entry, which are both important engines for economic growth. The latter two policy instruments are particularly effective at boosting growth in the short run. To confirm this argument, I replace the dependent variable with the share of language involving creating a healthy business environment in the city government's work reports, the log of per capita fixed investment, and the log of per capita fiscal expenditure, respectively, and estimate the baseline specification. The results show that having social ties with investigated officials significantly increases these three outcome variables. For example, the fixed investment and fiscal expenditure increase by almost 25%, implying that connected officials are desperately trying to boost economic growth for promotion using policy instruments. In particular, exploiting a unique firm registration data set, I find that firm entry significantly increases. Firm productivity also increases, especially for the firms that do not possess political connections with city leaders, indicating an improvement in the business environment.

If network-based political competition for promotion is the main driver of the empirical results, one would expect that the effects of the anti-corruption crackdown on connected officials should be larger when promotion incentives are more pronounced. This is the case when (1) the social group and, thus, the candidate pool for the job vacancy among city leaders are smaller (so that each member in the pool has a greater chance of being promoted)[6]; (2) the connected city official lags behind the promotion competition; (3) the time is approaching 2017 when a massive reshuffling of government officials took place; and (4) the official of interest works in a specialized ministry,[7] where the positions are persistently occupied by those from a certain educational background. I find that the effects are, indeed, stronger in these cases. Moreover, the empirical results are not fully consistent with other explanations, such as the frightening effects of anti-corruption efforts that induce officials to work harder (Wang, 2019).[8]

I also examine whether the impact of the anti-corruption shock will decay as the source of the shock becomes more distant in the workplace network. The results suggest that the effect of the anti-corruption shock dampens as the source of the shock becomes more distant in the network. Such results rule out the scenario in which anti-corruption shocks and economic performance are driven by confounding factors that affect both of them at the same time. Such evidence also points to the argument that social networks are key determinants of the effects of (anti)corruption and economic growth.

Although the crackdown on corrupt officials and the resulting political competition within social networks boost economic growth in some cities, they also incur costs to the economy. Focusing too much on short-term economic performance leads to the neglect of long-term development goals. Here, I focus on three outcomes: innovation, education expenditures, and pollutant emissions. The results indicate that the social ties with investigated officials significantly reduce the number of patent applications and R&D expenditure, the share of education expenditures (in total fiscal expenditures), and significantly increase the emission intensity (measured by the amount of pollutants normalized by GDP) of several pollutants. The city government also tends to overlook these goals in the government work reports. Therefore, intra-network competition may have long-run costs.

The rest of this paper is organized as follows. Section 2 conducts a literature review. Section 3 introduces the background and conceptual framework of this paper. Section 4 describes the data used in the empirical analysis. Section 5 presents the empirical results. Finally, Section 6 concludes.

2. Literature review

This paper speaks to six strands of the literature. First, this paper contributes to the literature on the political economy of economic growth or politicians' economic performance. Acemoglu et al. (2001, 2005) argue that institutions matter for economic growth, while Jones and Olken (2005), Besley et al. (2011), and Shi et al. (2020) believe that political leaders also matter. In this paper, I combine these two views together: I document that both the recent anti-corruption campaign, which can be seen as an institutional transition, and the local leaders, affect the economic growth rate at the city level. Moreover, Khan et al. (2019) argue that promotion incentives yielded by posting increase the economic performance of the politicians. Similarly, in this paper, I find that promotion incentives yielded by the creation of job vacancies following the anti-corruption crackdown matter for politicians' economic performance in China.

Second, this paper is related to the literature on city growth and development. Henderson and Wang (2007) argue that institutions are an important factor for city growth. The main takeaway argument in this paper echoes their argument since I find that the organization of politicians' social networks matters for the effects of anti-corruption on a city's economic growth. Furthermore, Da

[6] Note that this argument is equivalent to saying that regressing the GDP growth rate against the interaction term of the group size and a dummy of investigated social peers (and other controls and fixed effects as in the baseline specification) yields a negative regression coefficient on the interaction term.

[7] Such "network persistence" in ministries may stem from the requirement of specific skills for officials who work in these positions.

[8] In Wang (2019), the author does not investigate the role of social network in the transmission of shocks induced by anti-corruption investigation, and this is the major difference between that paper and this paper.

Mata et al. (2007) examine the determinants of city growth in Brazil. They state that city policies are core factors for city development. Social networks and factional politics, a core determinant for city leaders' incentives which also affect their policies. De Long and Shleifer (1993) and Bosker et al. (2008) both find that political regimes are essential for city growth, from a historical perspective. In this paper, I focus on a novel mechanism – political competition within social networks – underlying the effects of the anti-corruption campaign on city growth in China.

Third, it is related to the economic impacts of the (anti)corruption, especially in China. The main body of the literature on the effects of (anti)corruption on economic growth exploits cross-country data sets (Mo, 2001; d'Agostino et al., 2016; Gründler and Potrafke, 2019). City-level studies within a country are few. Moreover, as for the literature on the recent anti-corruption campaign, Chen et al. (2021), Ding et al. (2020), and Xu and Yano (2017) study the impacts of anti-corruption campaign on firms.[9] Among these papers, Chen et al. (2021) find an overall negative impact of the crackdown on firm productivity and entry rates. However, Ding et al. (2020) find that the stock market responded positively to the announcement of strong anti-corruption actions; Xu and Yano (2017) show that firms located in provinces with stronger anti-corruption efforts invested significantly more of their newly acquired funds in R&D and generated more patents. Shu and Cai (2017) find a negative effect of the anti-corruption campaign on the stock return of firms that produce liquor, which is widely used in political networking. Lan and Li (2018) and Qian and Wen (2015) explore the effects of the anti-corruption crackdown on the imports of luxury goods, and they both find a negative impact. Furthermore, Zhuang (2019) studies whether and how media is biased in reporting corrupt officials during the anti-corruption campaign. This paper is also closely related to Wang (2019), who finds that anti-corruption activities reduce the productivity of bureaucrats by frightening them away from informal practices. Thus, his conclusions are different from those in this paper. Like Wang (2019), I study the effects of the anti-corruption campaign on economic and policy outcomes at the regional level, and, thus, is distinct from most of the above papers. Moreover, I find that the economic impacts of the anti-corruption campaign by transmission in social networks are mixed: the campaign may boost the local economy at the expense of long-run welfare.

Fourth, this paper contributes to the studies on social ties and factional politics. Fisman et al. (2020), Jia et al. (2015), and Shih et al. (2012) study the effect of social connections on the selection and promotion of political leaders in China, and they have mixed findings. Benoit and Giannetti (2008), Boucek (2009), Brumfiel and Fox (2003), Chen and Hong (2016) Francois et al. (2016), Hong and Yang (2018), Huang (2006), Keller (2016), and Jiang (2018) conduct research on factional politics. Most importantly, Francois et al. (2016) identify two channels in which factions may affect promotion: on the one hand, a faction provides support to its members in obtaining promotion up the political hierarchy; on the other hand, a higher-ranked member of a faction may block the chance of promotion for junior members by higher ranked members, since the latter may want to avoid competing with them for future job openings. In this paper, the empirical results are established on the first channel: because of the support within a network, officials in the same social network as the dismissed corrupt official have lower costs for competing for the new job opening, and, thus, the promotion competition starts inside the network. Moreover, this paper is related to Lorentzen and Lu (2018), who study whether personal connections to high-ranked officials might protect one from anti-corruption investigations. Their finding that high-ranked officials will not provide protection for low-ranked ones is also supported by this paper.

Fifth, this paper studies promotion competition in China. In a seminal paper, Li and Zhou (2005) establish the positive relationship between economic performance and promotion for provincial officials in China. Yao and Zhang (2015) find that city leaders in China have significantly different levels of abilities to promote local economic growth and that personal abilities matter for promotion (as a leader gets older). Xu (2011) argues that China's institution can be regarded as a regionally decentralized authoritarian system, in which the central government has control over personnel, whereas local governments are responsible for running the economy. This line of research lays the foundation for the empirical results in this paper, in two respects: (1) since economic performance matters for promotion, they are expected to be better after the promotion competition begins. (2) since city officials have discretion over local affairs, economic and policy outcomes may vary as the officials' incentives change. As a result, it makes sense to investigate the outcomes of the anti-corruption crackdown.

Finally, this paper contributes to the research on the transmission of shocks in networks. Acemoglu et al. (2012) argue that intersectoral input–output networks might transmit microeconomic idiosyncratic shocks and lead to aggregate fluctuations. Acemoglu et al. (2015), Allen and Gale (2000), and Elliott et al. (2014) study how financial networks propagate shocks. Banerjee et al. (2013) examine how participation in a microfinance program diffuses through social networks, and they find that microfinance participation is higher when the people who were first informed about the program had higher eigenvector centrality. This paper relates to that literature by discussing how anti-corruption shocks affect social peers' economic performance and policy outcomes, or, in other words, how anti-corruption shocks are transmitted in the social networks of government officials.

3. Background

3.1. Institutional background

3.1.1. Social ties in China

Social ties, or in Chinese, *shehui guanxi*, play a pivotal role in the society, economy, and culture of China. Social ties are established through shared experiences or backgrounds; Social ties are often considered to include hometown ties (in Chinese, *tongxiang*),

[9] In addition, more broadly speaking, Manion (1996), Fisman and Wang (2015), and Chen (2022) study the implications of corruption on Chinese firms.

classmate ties (*tongxue*), and workplace ties (*tongshi*). These ties are pervasive in Chinese social contexts and lie at the heart of China's economic structure and changing institutional landscape. Indeed, social ties are important in almost every realm of life.

Social ties, or shared backgrounds, may define political factions. For example, Li (2014) documents that, the Shaanxi Gang, which is a group of officials who were born or have family origins in Shaanxi province, or, who have spent a large part of their career there, emerged at the power center of Beijing at the same time as Xi Jinping's ascent to the top leadership. Another example is graduates from Tsinghua University, which is the most commonly represented college within Politburo (Fisman et al., 2020).

In a non-democratic country such as China, social networks and factional politics play a very important role. The selection of government officials is one example where social or factional connections matter (Fisman et al., 2020, Francois et al., 2016, Jia et al., 2015, Shih et al., 2012). Social networks or factions may also affect the distribution of economic resources (Jiang and Zhang, 2020). Moreover, factions compete with one another in pursuit of power (Hong and Yang, 2018).

3.1.2. Network persistence

Network persistence is a phenomenon in which some positions in the political system are persistently occupied by certain networks (or factions). It is illustrated by Reiter (2004), and it is a distinguished feature of the factional politics in China. It is also the prerequisite of the analysis of this paper.

There are some statistics to illustrate the notion of network persistence, which indicates that some positions in the bureaucratic system in China are possessed by politicians in the same social network. The probability that any randomly chosen two city Party Secretaries who have ever worked in the same city were born in the same hometown province is 35.08%. The probability for city mayors, provincial Party Secretaries, and provincial governors who have worked in the same locality is 40.59%, 8.50%, and 15.1%, respectively. Since there are 34 provinces in China in total, the probability of randomly choosing two persons from these provinces is 2.94%. Thus, the probability that the two city Party Secretaries, city mayors, provincial Party Secretaries, and provincial governors having ever worked in the same locality were born in the same province is 11.93, 13.80, 2.89, and 5.14 times higher than that of randomly picking from 34 provinces, respectively.

Likewise, the probability of graduating from the same college for random two city Party Secretaries having ever worked in the same city is 3.81%. The probability for city mayors, provincial Party Secretaries, and provincial governors is 4.43%, 1.82%, and 3.82%, respectively. There are 279 different colleges in the sample and the probability of picking two officials randomly from the same college is 0.3%. Thus, the probability of graduating from the same college for city Party Secretaries, city mayors, provincial Party Secretaries, and provincial governors is 12.7, 14.8, 6.07, and 12.7 times higher than that of randomly picking from the 279 colleges, respectively.

Finally, the path of promotion and past work experience for any two bureaucrats in the same position is highly similar. The probability of having any overlap in the past work experience for any randomly chosen two city Party Secretaries, city mayors, provincial Party Secretaries, and provincial governors is 73.6%, 70.8%, 82.4%, and 80.1%, respectively. To sum up, all the above statistics point to the argument that network persistence is a prominent feature in the Chinese political system.

3.1.3. Intra-network support

Intra-network support is a phenomenon in which higher-ranked officials may help lower-ranked ones in the same network or faction to achieve promotion. It is illustrated by Francois et al. (2016), and there is anecdotal evidence supporting the existence of it in the Chinese bureaucratic system. The incentives justifying its existence is that when higher-ranked officials help lower-ranked ones to achieve promotion, the former are accumulating political capital—subordinates who are loyal to them and can help them achieve promotion in the future themselves.

As Shih (2016) argues, "lower-level officials join factions in order to secure promotions and other regime goods from powerful patrons". Along this line of reasoning, Francois et al. (2016) argue that "the advantage of social networks or factions is that they provide support to their members in obtaining promotions up the pyramid", and "factions allow the allocation of that support to be decided by senior affiliates". This is the theoretical foundation for intra-network support, which is another prerequisite for the empirical analysis in this paper.

3.1.4. Economic performance and promotion in China

China's political system has five levels of state administration: the central government, provincial governments, prefecture city governments, county governments, and townships. The Central Committee of the Chinese Communist Party (CCP) acts as the headquarter of this multi-divisional system and has final control over the appointment of government officials at the lower levels within the system. Prefecture cities, the focus of this paper, are the third level of China's political hierarchy. The top position at the prefecture city level is the city party secretary, and the second-highest position is the city mayor. This reflects the duality of the CCP and government organs at each level of China's political hierarchy (Li and Zhou, 2005).

China started its economic reforms in 1978. These reforms allowed local governments to play a much more pivotal role in economic development than the ministries at the central government, which were previously in charge of planning and coordination of the economy. Such reforms reveal the significance of city leaders, including the city's Party Secretary and the city mayor. In addition, reforms have also given city leaders the authority to manage economic affairs in their cities. As a result, the economic performance of the cities is largely affected by the decisions of the city leaders. For this reason, the city leaders are also held accountable for the results of their decisions.

China's reforms of its political selection system coincided with the beginning of its economic reforms. An important turnabout in personnel management was the change in the evaluation system of government officials. Loyalty, which was the only crucial criterion for promotion before the reform, gave way to economic performance. Officials had to be young, well-educated, and good at administrative management. All said, local economic performance became the most important criterion by which higher-level officials evaluated lower-level ones (Li and Zhou, 2005).

3.1.5. Anti-corruption campaign in China

Anti-corruption has been an important policy issue of successive Chinese governments. Since the economic reforms began in 1978, corruption in China has emerged rapidly and widely. There are many types of corruption, though they usually involve trading graft for economic and political favors, such as local businesspeople trying to win large government projects or lower-level officials trying to seek promotion to higher-level jobs.

In 2013, which is the starting year of my sample, China ranked 80th of 177 countries and regions in the Corruption Perceptions Index reported by Transparency International. Such a high level of corruption raised serious concern among Chinese top leaders. General Party Secretary Xi Jinping told the Politburo in November 2012: "If corruption becomes increasingly rampant, it will inevitably lead to the downfall of the Party and the State.[10]" After this statement (and the end of the 18th National Congress of the Communist Party in 2012) the intensive anti-corruption campaign began, and it yielded significant impacts on Chinese politics. Xi promised to punish "tigers and flies"-i.e., high-level officials and grassroots alike. Most of the officials investigated were removed from office and penalized. As of now, the campaign has involved over 120 high-ranking officials, including former Politburo Standing Committee (PSC) member Zhou Yongkang and former military leaders Xu Caihou and Guo Boxiong.

3.2. Conceptual framework

The existing literature has mixed findings regarding the effects of (anti-)corruption on economic growth. Mo (2001) exploits a cross-country data set and finds that corruption has negative impacts on economic growth at the national level. Such a finding is further corroborated by Gründler and Potrafke (2019). However, in the case of China, Qu et al. (2018) find that anti-corruption in China has negative effects on economic growth in Chinese provinces. While these papers estimate the total effects of (anti-)corruption on growth, I focus on the effects through one specific channel: network-based political competition,[11] while admitting that there may be other channels as well. I will now illustrate potential hypotheses that describe how this specific channel is at play.

Having social ties with officials who have been investigated for corruption may influence the economic performance of a city's mayor or Party Secretary in three ways. First, the investigation, and the subsequent dismissal of officials, might lead to new job openings for all other officials. As a result, there might be promotion competition among officials who have not been investigated for corruption. Driven by promotion incentives, they may devote more effort to boosting economic growth.

Moreover, such promotion competition is more likely to take place within a social network, which is defined by shared backgrounds in birthplace, education, and career paths. This is because higher-level officials might support lower-level ones in the same social network in obtaining promotions in the bureaucratic hierarchy (Francois et al., 2016). In addition, some positions may be controlled by specific networks, and, thus, the succession happens within a network or a faction, such as officials from the same school occupy the positions of some specific fields of ministry, resulting in network-based "persistence" or "balance" (Reiter, 2004). Therefore, the switching costs to new jobs for officials within the social networks are smaller than those for officials outside the faction; hence, an official socially connected to the dismissed corrupt peers might have stronger promotion incentives in the competition. In response to such incentives, the officials will devote more effort to boosting local economic growth. This mechanism is the intra-network competition.

Suppose there are two cities, A and B. The leader in A has a boss, and also a social connection, who is arrested for corruption and, thus, removed from his job. Ceteris paribus, the chance of promotion for the leader is A is larger, and, thus, A has a stronger incentive to work harder to compete for his boss's position. As a result, his economic performance increases.

The second way in which anti-corruption crackdown on social peers might affect a city official's economic performance is that the connected city official may receive greater attention from the anti-corruption agency and, thus, they are not willing to be promoted because they would have to undergo a thorough audit of personal property, which may reveal hidden corruption. As a result, being connected to corrupt peers may worsen economic performance. This mechanism is the spillover of anti-corruption attention.

Finally, the investigation and punishment of a government official might be a form of political purge and, thus, it cracks down on the entire social network or political faction. The new job opening after the removal of the corrupt official would only be available for members of a new network. As a result, the investigation of a corrupt official reduces the promotion incentives for all members in the same social network, and the performance of these officials would be worse. Actually, the mechanism of intra-network competition is a new mechanism that has not been intensively discussed in the existing literature. This is where the main contribution of this paper stands. To summarize these mechanisms, we have Hypotheses 1 and 2.

Hypothesis 1. When the mechanism of intra-network competition is at play, for a city official socially connected to other officials who have been investigated for corruption, the economic performance of that official becomes better.

Hypothesis 2. When the mechanism of the spillover of anti-corruption attention or the purge of the same social network or political faction is at play, being connected to investigated officials might make the economic performance worse.

[10] Bloomberg News. https://www.bloomberg.com/news/articles/2013-12-30/china-s-xi-amassing-most-power-since-deng-raises-risk-for-reform.

[11] Using the investigation of connected political leaders enables me to estimate the effect of this channel.

Next, consider the comparative statics, which are specific to different mechanisms. When intra-network competition is the main driver of the correlation between having social ties with corrupt peers and economic growth, the partial effects of social ties should be larger, as the promotion incentives are more pronounced (for example, as time of the massive reshuffling of government officials approaches). Moreover, when the diffusion of anti-corruption attention is at work, the magnitude of the effects of social ties should increase, as the city official's social group attracts more attention from the anti-corruption agency (for example, when there are more corrupt officials in the same social group). These points are what Hypotheses 3 and 4 capture.

Hypothesis 3. When the mechanism of intra-network competition is at play, the effects on the economic performance of social ties with officials who have been investigated for corruption are more positive when promotion incentives are more salient.

Hypothesis 4. When the mechanism of the spillover of anti-corruption attention is at play, such effects become more negative as the attention from the anti-corruption agency is greater.

Finally, consider the costs of the intra-network competition that results from the anti-corruption crackdown. After the dismissal of officials in the same social network, if the remaining city officials want to win the promotion competition for the new job openings, then they must signal their capacity by enhancing economic growth, which is an important criterion for the evaluation of the job candidates. As a result, city officials will rely on policy instruments such as investment and public spending to boost growth in the short run. However, such a strategy works at the cost of long-term development. Consequently, development goals such as innovation, education, and environmental protection might be overlooked. This leads to Hypothesis 5.

Hypothesis 5. If the mechanism of intra-network competition is at play, and a city official has social ties with officials who have been investigated for corruption, then, the goals of long-term development in that city, such as innovation, education, and the environment, are compromised.

In the Appendix, I build a parsimonious model to rationalize these hypotheses, especially Hypotheses 1, 3, and 5.

4. Data

The data are from five sources. The first source is the China Center for Economic Research official data set, which provides detailed personal and career information on all officials above the prefecture-city level, especially city mayors and city party secretaries,[12] working during 2013–2017.[13] Although my regression sample covers city-level officials, I use all officials in the data set to construct measures on social networks or factional linkages. This is a new data set that contains comprehensive information on the social networks of major above-middle-ranked bureaucrats.

The second data source is the list of the names of all investigated officials, provided by the website of the Central Commission for Discipline Inspection (CCDI, www.ccdi.gov.cn) and China Judgements Online I matched the information on investigations with the China Center for Economic Research data set on officials. Since the investigation list encompasses all levels of the bureaucratic hierarchy, the matched officials are a subsample of all investigated officials.

During the anti-corruption campaign in China, the investigations were all conducted with strong evidence that the investigated officials already violated laws or the discipline of the Party. No investigations that were publicly announced were conducted without noticeable violations. As a result, the investigated officials were almost surely punished later and were dismissed from their current positions.

The data set used in the regression analysis is official-year (unbalanced) panel data, containing 935 officials (mentioned above), 287 cities, and five years (from 2013 through 2017). The data set has time variations in officials' corruption investigations and social ties.

The CCDI and China Judgements Online websites contain information on the timing of the corruption investigation, and, accordingly, I construct the independent variables based on whether the official has social ties with other officials who have been investigated for corruption. To alleviate endogeneity, I use the connections established prior to being in office as the measure for social connections, since they are predetermined. Consider, first, my measure based on shared hometown. I define candidate i for the year t to be hometown-city-connected with officials who have been investigated for corruption ($CityTie_{it} = 1$) if there exists at least one investigated official at t who is from the same prefecture city as i.

I similarly construct *College Tie* based on officials' undergraduate institutions. (For officials without a college degree, we set $CollegeTie_{it} = 0$. Removing all officials without a college degree from the sample yields qualitatively similar results.) In the regressions using college ties as the independent variable, I control for education levels in order to avoid conflating the role of shared background with educational attainment.

For shared work background, I require that ($WorkTie_{it} = 1$) when official i and investigated officials have a period of overlap in their work histories, a period of time in which both worked in the same organization/department in the same prefecture (when the

[12] The research sample consists of both city Party Secretaries and city mayors who have approximately equal sample sizes.

[13] The primary reason to use data up to 2017 is that the main database that provides information on officials' social networks, the CCER official database, only has data till 2017. Moreover, according to the Baidu Index, the topic popularity of anti-corruption decreased from 1335 in December 2012 (when the anti-corruption campaign started) to 340 in December 2017, to 292 in December 2018, and further to 172 in January 2020. Thus, from the perspectives of the public, anti-corruption has become a less prominent policy issue after 2017.

official is at the prefecture level, whereas the workplace ties are defined according to working in the same provincial governmental agency if the official is at the provincial level.).

Moreover, on the CCDI website, I obtained information regarding the objectives and the timing of inspections by the Central Inspection Team. According to such information, I construct three dummy variables on whether an official's city ties, college ties, and work ties have ever been investigated by the Central Inspection Team. These variables are used as the instruments in the empirical analysis.

The third data source is the granular firm-level databases. I use a unique firm registration data set to measure firm entry. This data is also used in Shi (2021c), Shi et al. (2022), Shi and Wang (2022). I also use a data set from the annual waves of the National Tax Statistics Database (NTSD), which are jointly conducted by the State Administration of Taxation and the Ministry of Finance. This data set provides information on firm productivity and innovation activities, including R&D expenditures.

The fourth data source is the government work reports of all cities during 2012–2017. I extract keywords concerning business environment, innovation, education, and environmental protection, and calculate its share in the total report. I use these shares as the dependent variable to measure city leaders' policy choices.

The fifth data source is the China City Statistical Yearbooks. The Yearbooks have annual data on the economic growth and other characteristics of prefecture cities in China.

The summary statistics of the officials in the sample are shown in Table A.1. About 21.3% of the observations have city ties with officials who have been investigated for corruption, while the percentage for college ties and workplace ties are 8.4% and 4.4%, respectively. The mean of the main dependent variable, per capita GDP growth rate, is about 8.99%.

5. Empirical results

5.1. Baseline results

My main analyses investigate the relationship between social ties with investigated officials and economic performance, including a range of controls and fixed effects. I use a city-year-official level longitudinal data set and a generalized difference-in-differences strategy. The specification takes the following form:

$$y_{ilt} = \alpha 1(SocialTie)^c_{it} + X_{1,it}\beta_1 + X_{2,lt}\beta_2 + \gamma_c + \omega_t + \lambda_l + u_{ilt}, \tag{1}$$

where y_{ilt} is the outcome variable for official i, city l, and year t. $1(SocialTie)^c_{it}$ is an indicator variable equal to 1 if official i has one of the social ties (city tie, college tie, workplace tie) to any official who has been investigated for corruption at the time t. So α is the parameter of interest. The anti-corruption campaign can be seen as a quasi-natural experiment. The anti-corruption shock, due to its secrecy beforehand and abruptness of implementation, can be seen as exogenous variation relative to the economic performance and policy outcomes, especially conditional on city-, network-, and year-specific fixed effects that absorb all city-, network-, and year-specific factors that may determine the objective of the investigation. Thus, α is well-identified (in the sense that there is mean independence between the error term and the covariates: $E(u|1(SocialTie), X, \gamma, \omega, \lambda) = 0$). $X_{1,it}$ is a vector of official-level control variables, including the age, education, ethnicity, and gender of official i. $X_{2,lt}$ is a vector of city-level control variables, including the lag of log population and the lag of GDP share of the secondary industry in city l. γ_c is one of the social group fixed effects (city FE, college FE, workplace FE). It absorbs all differences across social groups. ω_t is the year fixed effects, which include the impacts of central inspection teams on all officials. λ_l is the city fixed effects, which control for all time-invariant unobserved city-level heterogeneity. Therefore, the variations used to identify the effects of connections on GDP growth are within-city, within-year, and within-network variations. For robustness checks, I also include official fixed effects and, hence, use within-official variations for identification. The standard errors are clustered at the city level.

I present the main OLS (generalized difference-in-differences) results in Table 1. In columns (1) through (3), in which I use *City Tie* as our connection measure and include hometown city fixed effects, the per capita GDP growth rates for connected officials are higher, but such effect is statistically insignificant.[14] In columns (4) through (6), I use *College Tie* as our measure of connections. I find a positive impact of college ties on economic growth rates, which is about 2.5–2.8 percentage points (significant at least at the five-percent level). In columns (7) through (9), with *Work Tie* as the connection measure and workplace fixed effects, I again find a positive and statistically significant effect: connected officials have per capita GDP growth rates of about 2.3–2.6 percentage points higher. Using night-light intensity as the dependent variable to measure economic growth yields qualitatively similar results, as reported in Table B1. I also control for official-level individual fixed effects in the regressions. I do not control for such fixed effects in all regressions because there are few within individual variations to identify the causal impact of the investigations of social peers on economic performance, leading to a high level of multicollinearity. The estimation results are reported in Table B2, and are qualitatively similar to the baseline results. Finally, I also add social group variables, including the size and the proportion of corrupt officials in the social group, as control variables, and the results, reported in Table B3, are also qualitatively similar to the baseline results. All of these results are consistent with Hypothesis 1, not Hypothesis 2.

I then augment the estimation using an instrumental variable approach. I instrument the social tie variables with a dummy indicating whether the corresponding social ties have been inspected by the Central Inspection Team (CIT). This approach is based

[14] The fact that city ties do not have significant impacts may result from the observation that hometown connections are "too strong" in the sense that politicians with hometown connections may not have higher probability of being promoted, for the sake of avoiding suspicion of cronyism or corruption.

Table 1
Baseline results.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	per capita GDP growth rates (%)								
1(having city tie to investigated officials)	0.689	1.245	0.382						
	(0.926)	(1.045)	(1.308)						
1(having college tie to investigated officials)				2.861***	2.745***	2.555**			
				(1.005)	(1.025)	(1.014)			
1(having workplace tie to investigated officials)							2.382**	2.685*	2.477*
							(1.134)	(1.449)	(1.389)
Home City FE	Y	Y	Y	N	N	N	N	N	N
College FE	N	N	N	Y	Y	Y	N	N	N
Workplace FE	N	N	N	N	N	N	Y	Y	Y
City FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Official Controls	N	Y	Y	N	Y	Y	N	Y	Y
City Controls	N	N	Y	N	N	Y	N	N	Y
Observations	2,753	2,530	2,489	2,803	2,553	2,511	2,803	2,553	2,511
R-squared	0.393	0.400	0.448	0.328	0.333	0.380	0.296	0.300	0.335

Notes: The sample covers 935 officials, 287 cities, and 5 years from 2013 to 2017. Official controls include the age, a dummy for college and a dummy for graduate school, ethnicity dummies, and a gender dummy of the officials. City controls include the lag of log population and the lag of GDP share of secondary industry. * Significant at 10%, ** 5%, *** 1%. The standard errors are clustered at the city level.

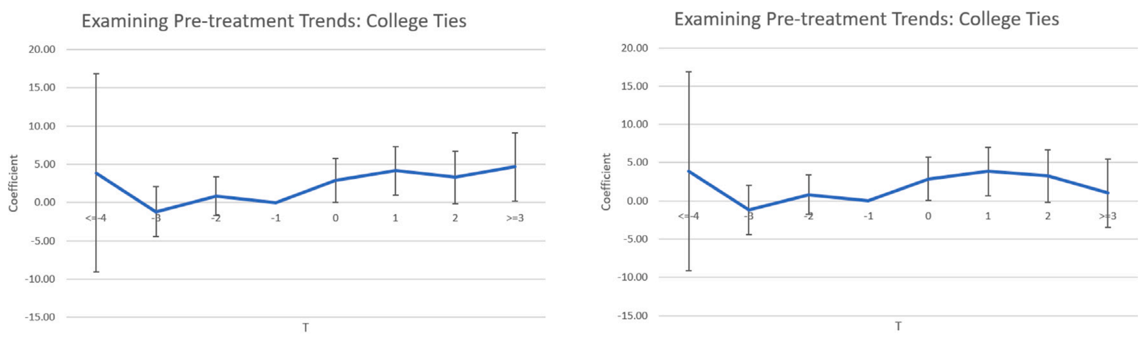
upon an assumption that the objective of inspection by CIT is randomly chosen conditional on the controls and fixed effects, especially the year fixed effects regarding the decisions of the central government, and the social group fixed effects that contain all time-invariant characteristics that attract the attention from the CIT.[15] Table A2 reports the results of the IV estimation and the associated first-stage regressions. In columns (1) through (3), the coefficients on the social tie variables are all positive, although for city ties, the coefficient is not precisely estimated. The results for the IV estimation are qualitatively similar to those of the OLS estimation, but the magnitude of the coefficients for IV estimation is somewhat larger than those for the OLS estimation. The Kleibergen–Paap rk Wald F statistics imply that there is no weak instrument issue. Columns (4) through (6) show the results for the first-stage regressions. All regressions show a significant and positive correlation between the social tie variables and the instruments.

Based on the existing literature, the usual results are that researchers find the 2SLS estimate to be larger than the OLS estimate by approximately 1.25 to 10 times, e.g. (Card, 1999; Card, 2001). There are three reasons why the IV estimates are larger than the OLS estimates: (1) An omitted variable that could be negatively correlated with the investigation of the connected officials, for example, the unobserved ability of politicians. This omitted variable would lead to a downward bias in the OLS estimate of the connection coefficient. (2) Errors in variables that arise because observed college and workplace ties are a noisy measure of real social relationships that lead to better economic performance. This measurement error in connection biases the OLS estimate of the treatment effect toward zero. Since the IV estimate is unaffected by the measurement error, they tend to be larger than the OLS estimates. (3) It is possible that the IV estimate is larger than the OLS estimate because IV is estimating the local average treatment effect (LATE), while OLS is estimating the ATE over the entire population. IV is estimating the LATE: the instrument shifts the behavior of a subgroup of individuals for whom the connection dummy equals one. In other words, the IV estimate is the effect of gaining connection only for the population whose choice of treatment was affected by the instrument, while the OLS estimate describes the average difference in economic performance for those whose connection dummy differs by one. Then IV estimates will be larger than OLS estimates because of heterogeneity in the studied population.

I then take an event-study approach to estimate the dynamic effects of social ties with investigated officials on economic performance. The results are presented in Fig. 1 (for college ties) and Fig. 2 (for work ties). In subfigure (a), the time is calculated as the number of years since a city is hit by an anti-corruption shock, conditional on no turnover has ever happened since then. In subfigure (b), the time is calculated as the number of years since a city is hit by an anti-corruption shock, allowing for turnover to happen since then. The results indicate that there is no unobserved pre-treatment trend in the "treated" group (those who are connected to any investigated officials). The subfigure (a) in these figures show that the effects of anti-corruption investigations are quite persistent. This may be due to the fact that the officials compete not only directly for the job openings previously held by the removed officials but also for the job openings that are created by the officials who fill the original job openings, thus leaving a long chain reaction after the removal of the original corrupt officials. Moreover, since there has not been turnover since $t = 0$, the city leader may become more desperate for promotion over time and, thus, the coefficients are inflated. If I allow for turnover to happen, as in subfigure (b), then the coefficients become smaller. This is because once turnover happens after $t = 0$, there is no such persistent effect of anti-corruption shock as above.

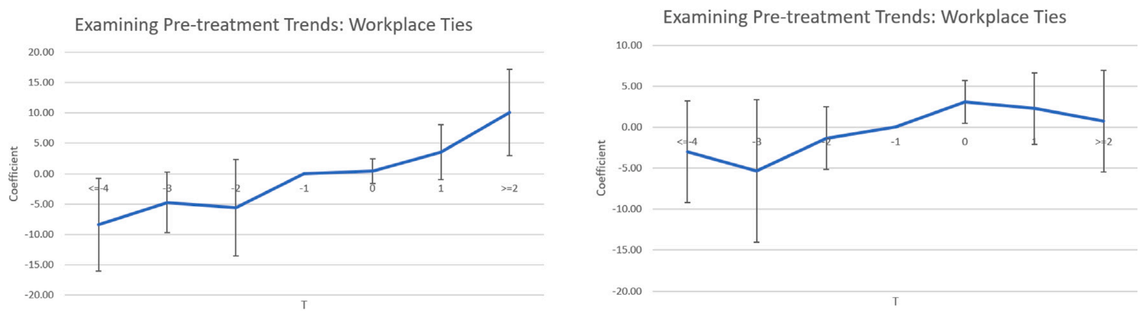
In addition, I test whether other types of removal of connected bureaucrats that produce job vacancies, such as retirement, can also provide promotion incentives and induce city leaders to work harder. I construct the main independent variables as to whether there is any job vacancy created by retiring connected officials and rerun the baseline specification. The results are reported in

[15] The instrument may not be valid without these controls.



(a) (b)

Fig. 1. Estimating the dynamic effects of college ties on Per capita GDP growth rate.
Notes: The figure illustrates the dynamic effects (three-year range before and three-year range after the treatment) of the treatment $1(t=\tau)\times 1(CollegeTie_i)$ on per capita GDP growth rate. The horizontal axis measures the year since the official experienced a treatment. 0 represents the first year of the treatment. The vertical axis measures the regression coefficients for the dynamic effects. The coefficients are obtained using the baseline specification (with controls, and various fixed effects), with the only exception that the dummy for the treatment is replaced by the interaction terms of the dummy for treatment and dummies for time. In subfigure (a), the time is calculated as the number of years since a city is hit by an anti-corruption shock, conditional on no turnover has ever happened since then. In subfigure (b), the time is calculated as the number of years since a city is hit by an anti-corruption shock, allowing for turnover to happen since then. The figure shows that the positive effect of the college ties is restricted to the period in which the treatment has started ($t \geq 0$). The vertical line around each plotted coefficient indicates the 95% confidence interval, with standard errors clustered at the official level. Every estimated effect is compared to the year that is one year prior to that of the treatment, which is standardized to 0.



(a) (b)

Fig. 2. Estimating the dynamic effects of work ties on Per capita GDP growth rate.
Notes: The figure illustrates the dynamic effects (three-year range before and two-year range after the treatment) of the treatment $1(t=\tau)\times 1(WorkTie_i)$ on per capita GDP growth rate. The horizontal axis measures the year since the official experienced a treatment. 0 represents the first year of the treatment. The vertical axis measures the regression coefficients for the dynamic effects. The coefficients are obtained using the baseline specification (with controls, and various fixed effects), with the only exception that the dummy for the treatment is replaced by the interaction terms of the dummy for treatment and dummies for time. In subfigure (a), the time is calculated as the number of years since a city is hit by an anti-corruption shock, conditional on no turnover has ever happened since then. In subfigure (b), the time is calculated as the number of years since a city is hit by an anti-corruption shock, allowing for turnover to happen since then. The figure shows that the positive effect of the work ties is restricted to the period in which the treatment has started ($t \geq 0$). The vertical line around each plotted coefficient indicates the 95% confidence interval, with standard errors clustered at the official level. Every estimated effect is compared to the year that is one year prior to that of the treatment, which is standardized to 0.

Table B4. They indicate that this is exactly the case. However, such results must be interpreted with caution since the retirement of bureaucrats is not an exogenous event. On the other hand, if there are other officials within the same province who are investigated, then the economic performance of other uninvestigated officials within the same province remains unaffected, as Table B5 suggests.

The result that the anti-corruption campaign may have positive, although indirect, effects on the economy contradict the widely embraced view that the anti-corruption campaign may harm the Chinese economy.[16] I argue that such positive effects may stem from political incentives created by removing current corrupt officials from their jobs: the anti-corruption investigation of officials initiates promotion competition inside the social network, which is defined by shared backgrounds.[17] Therefore, this paper has a

16 Such arguments are common in Chinese and Western media. For example: https://chinapower.csis.org/china-corruption-development/.

17 Being connected to an investigated official, especially one with a higher rank, provides a promotion opportunity to the rest. I test whether social ties with an investigated official will lead to a better chance of being promoted, and the answer is yes. Moreover, I find that economic performance will contribute more to promotion if the candidate is connected to an investigated official.

Table 2
Anti-corruption shocks, GDP growth, and promotion.

	(1)	(2)	(3)	(4)	(5)	(6)
	1(Promoted next year)					
	OLS		IV-2SLS			
1(having college tie to investigated officials)	0.0605*		1.382***			
	(0.0322)		(0.316)			
1(having workplace tie to investigated officials)		0.0609		0.485**		
		(0.0610)		(0.192)		
Per capita GDP growth ranking					−0.0613	0.0323***
					(0.0387)	(0.0111)
Per capita GDP growth ranking*1(college tie)					1.859**	
					(0.723)	
Per capita GDP growth ranking*1(workplace tie)						0.764*
						(0.431)
Kleibergen-Paap rk Wald F statistic			19.582	5.336	19.582	5.336
Year FE	Y	Y	Y	Y	Y	Y
Official FE	Y	Y	Y	Y	Y	Y
Observations	2,835	2,835	2,835	2,835	2,780	2,780
R-squared	0.402	0.402	0.166	0.397	0.405	0.398

Notes: The sample covers 935 officials, 287 cities, and 5 years from 2013 to 2017. The instrumental variables in the IV-2SLS regressions in columns (3), (4), (5), and (6) are a dummy variable indicating whether the college ties and work ties have been inspected by CCDI. Per capita GDP growth ranking is defined as $Ranking = \frac{g_{ipt}-mean(g_{ipt})}{sd(g_{ipt})}$, where g_{ipt} is the growth rate of city i in province p and year t; $mean(g_{ipt})$ is the mean of all cities' growth rates in province p; and $sd(g_{ipt})$ is the standard deviation of all cities' growth rates in province p. * Significant at 10%, ** 5%, *** 1%. The standard errors are clustered at the city level.

policy implication that the structure of politicians' social networks, matters for the economic development of the country (Jiang et al., 2022b; Jiang et al., 2022a).

A natural follow-up question is whether the anti-corruption shocks may actually create promotion opportunities for the officials connected to the investigated peers. To consider this question, I estimate a promotion equation in which the outcome variable is whether the official is promoted in the next year, and the independent variable is whether the official has any social connection to investigated peers. Table 2 presents the results. Columns (1) and (2) show the estimates for college ties and work ties for the OLS estimation, and columns (3) and (4) show the results for the IV-2SLS estimation. All the coefficients on the social ties variables are positive, whereas the magnitude of the IV-2SLS estimates is larger, and the statistical significance is higher. Moreover, the GDP growth ranking among all cities in the province contributes positively to promotion for city-level officials.[18] In all, if an official's social connections are investigated and subsequently removed from job, and if that official has better economic performance, then that official has a better opportunity of being promoted. Such results are the premise of the whole story of this paper.

The next question is: how do officials achieve the goal of economic growth? The answer is that officials can boost economic growth in three ways: encouraging firm entry, attracting investment, and increasing government spending, as discussed in the literature: Shi (2021c), Shi et al. (2022), Shi and Wang (2022) argue that firm entry is a major source of economic growth in China; Levine and Renelt (1992) find that the share of investment is positively correlated with the economic growth rate; Barro (1990) establishes that the government can promote economic growth through government spending. To encourage firm entry, the city government may create a healthier business environment, so that it is more profitable for firms to enter the market (Shi, 2021a; Liu and Shi, 2023). I use the share of language concerning improving the business environment in the government work report, log firm entry, log of per capita fixed investment, and the log of per capita fiscal expenditure as the dependent variables, and test whether having social ties with officials being investigated for corruption has any impact. The results in Table 3 indicate that being connected to investigated officials significantly improves the share of language involving a healthy business environment in the government work report. Furthermore, in Table B6, I document that such connections increase firm entry, which may be a result of a healthier business environment.[19] Another result of such improvement is the boost of firm labor productivity, as is indicated by Table A3. In particular, the improvement is smaller for firms with political connections,[20] which is evidence for an improvement of the business environment. In Table 4, the results show that college ties increase per capita fixed investment by around 20%, while workplace ties increase it by about 10%. Moreover, Table 5 indicates that college ties and workplace ties raise per capita fiscal expenditures by approximately 20%.[21]

[18] Per capita GDP growth ranking is defined as $Ranking = \frac{g_{ipt}-mean(g_{ipt})}{sd(g_{ipt})}$, where g_{ipt} is the growth rate of city i in province p and year t; $mean(g_{ipt})$ is the mean of all cities' growth rates in province p; and $sd(g_{ipt})$ is the standard deviation of all cities' growth rates in province p.

[19] The previous literature, such as Shi et al. (2021) and Liu and Shi (2023), argues that political connections between politicians and businesspeople may induce corruption and rent-seeking opportunities, and incur a social cost to the economy. Therefore, a healthier business environment is consistent with the fact that the performance of connected firms improves relatively less.

[20] The definition of political connections is consistent with Shi et al. (2021), Shi (2021a). Using firm registration data, I define a firm to be politically connected (to the city leader) if the origin of its legal representative is the previous workplace or hometown of the city leader.

[21] The magnitude of the positive effects of connected investigations on firm entry, fixed investments, and fiscal expenditure is large, based on some quantitative estimates in the existing literature (Barro, 1990; Levine and Renelt, 1992; Shi et al., 2021) which reveals the intense short-term responses of city leaders to connected investigations.

Table 3
Effects on business environment.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Business environment improving in government report								
1(having city tie to corrupt officials)	0.0101***	0.0101***	0.0101***						
	(0.000251)	(0.000280)	(0.000285)						
1(having college tie to corrupt officials)				0.0127***	0.0127***	0.0130***			
				(0.000760)	(0.000786)	(0.000831)			
1(having workplace tie to corrupt officials)							0.0172***	0.0169***	0.0168***
							(0.00122)	(0.00110)	(0.00110)
Home City FE	Y	Y	Y	N	N	N	N	N	N
College FE	N	N	N	Y	Y	Y	N	N	N
Workplace FE	N	N	N	N	N	N	Y	Y	Y
City FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Official Controls	N	Y	Y	N	Y	Y	N	Y	Y
City Controls	N	N	Y	N	N	Y	N	N	Y
Observations	2,764	2,534	2,478	2,823	2,565	2,506	2,834	2,566	2,505
R-squared	0.817	0.822	0.824	0.568	0.603	0.608	0.552	0.576	0.579

Notes: The sample covers 935 officials, 287 cities, and 5 years from 2013 to 2017. Official controls include the age, a dummy for college and a dummy for graduate school, ethnicity dummies, and a gender dummy of the officials. City controls include the lag of log population and the lag of GDP share of secondary industry. * Significant at 10%, ** 5%, *** 1%. The standard errors are clustered at the city level.

Table 4
The effects of corruption investigation on fixed investment.

	(1)	(2)	(3)	(4)	(5)	(6)
	log(fixed investment per capita)					
1(having college tie to investigated officials)	0.210***	0.203***	0.218***			
	(0.0741)	(0.0757)	(0.0664)			
1(having workplace tie to investigated officials)				0.160*	0.0932	0.0932
				(0.0878)	(0.0912)	(0.0912)
College FE	Y	Y	Y	N	N	N
Workplace FE	N	N	N	Y	Y	Y
City FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Official Controls	N	Y	Y	N	Y	Y
City Controls	N	N	Y	N	N	Y
Observations	2,570	2,344	2,338	2,570	2,338	2,338
R-squared	0.818	0.834	0.862	0.803	0.859	0.859

Notes: The sample covers 935 officials, 287 cities, and 5 years from 2013 to 2017. Official controls include the age, a dummy for college and a dummy for graduate school, ethnicity dummies, and a gender dummy of the officials. City controls include the lag of log population and the lag of GDP share of secondary industry. * Significant at 10%, ** 5%, *** 1%. The standard errors are clustered at the city level.

Table 5
The effects of corruption investigation on fiscal expenditure.

	(1)	(2)	(3)	(4)	(5)	(6)
	log(fiscal expenditure per capita)					
1(having college tie to investigated officials)	0.234***	0.233***	0.259***			
	(0.0404)	(0.0420)	(0.0283)			
1(having workplace tie to investigated officials)				0.246***	0.241***	0.218***
				(0.0344)	(0.0393)	(0.0298)
College FE	Y	Y	Y	N	N	N
Workplace FE	N	N	N	Y	Y	Y
City FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Official Controls	N	Y	Y	N	Y	Y
City Controls	N	N	Y	N	N	Y
Observations	2,786	2,537	2,518	2,786	2,537	2,518
R-squared	0.834	0.834	0.952	0.824	0.826	0.944

Notes: The sample covers 935 officials, 287 cities, and 5 years from 2013 to 2017. Official controls include the age, a dummy for college and a dummy for graduate school, ethnicity dummies, and a gender dummy of the officials. City controls include the lag of log population and the lag of GDP share of secondary industry. * Significant at 10%, ** 5%, *** 1%. The standard errors are clustered at the city level.

Table 6
Network distance and shock transmission.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	per capita GDP growth rates (%)			log(fixed investment per capita)			log(fiscal expenditure per capita)		
1(Workplace ties, distance = 1)	2.477*			0.160*			0.218***		
	(1.389)			(0.0878)			(0.0298)		
1(Workplace ties, distance = 2)		1.464			0.0569***			0.0164	
		(1.090)			(0.0211)			(0.0102)	
1(Workplace ties, distance = 3)			0.285			−0.0273			0.000570
			(0.867)			(0.0216)			(0.0114)
Workplace FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
City FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Official Controls	Y	Y	Y	Y	Y	Y	Y	Y	Y
City Controls	Y	Y	Y	Y	Y	Y	Y	Y	Y
Observations	2,511	2,511	2,511	2,570	2,570	2,570	2,518	2,518	2,518
R-squared	0.335	0.336	0.335	0.803	0.803	0.802	0.944	0.943	0.943

Notes: The sample covers 935 officials, 287 cities, and 5 years from 2013 to 2017. Official controls include the age, a dummy for college and a dummy for graduate school, ethnicity dummies, and a gender dummy of the officials. City controls include the lag of log population and the lag of GDP share of secondary industry. * Significant at 10%, ** 5%, *** 1%. The standard errors are clustered at the city level.

5.2. Network distance and shock transmission

In this subsection, I test how anti-corruption shocks are specifically transmitted in politicians' social networks. Since the college network is a complete network in which the distance of all nodes is one, I focus on the workplace network, which is not a complete network, and in which the distance of nodes can be strictly greater than one.

I construct three variables: (1) 1(Workplace ties, distance = 1); (2) 1(Workplace ties, distance = 2); and (3) 1(Workplace ties, distance = 3), which indicates whether there is an anti-corruption shock whose distance of source is one, two, or three, respectively. Then, I use each of these three variables as the main independent variable and estimate the baseline specification, Eq. (1).

The results are reported in Table 6. In columns (1), (4), and (7), in which the distance of the source is one, the anti-corruption shock has positive effects on per capita GDP growth, per capita fixed investments, and per capita fiscal expenditures. However, in columns (2), (5), (8), (3), (6), and (9), as the distance becomes two or three, the magnitude of the effect of the anti-corruption shocks becomes much smaller. Thus, the results suggest that the anti-corruption shocks decay as the source becomes more distant. Such results rule out the scenario that the statistically significant correlation between anti-corruption shocks and outcomes might be driven by common confounding factors.

These results also point to the argument that the structure of local politicians' social networks matters for their performance and, thus, the outcome of economic growth. This is a viewpoint that has appeared in the previous literature (Jiang et al., 2022b; Liu and Shi, 2023).

5.3. Mechanism: intra-network competition

One explanation for the positive correlation between economic performance and social ties with investigated officials is that the investigation and the following punishment and dismissal of officials lead to a new open position and, thus, initiate promotion competition within the social networks of those officials. To test this explanation, I first examine whether there is indeed competition among officials in the same network. Using a city-dyad investment flow data set as in Shi (2021a) and Liu and Shi (2023), I test whether city leaders hit by the anti-corruption shocks in the same network will reduce cooperation with each other by reducing intercity investment flows. The results reported in Table B7 show that this is exactly the case. When a city leader has a social connection who has been investigated for corruption, that leader has a stronger promotion incentive and, thus, chooses not to cooperate with his rivals, or other city leaders in the same network. This is the direct evidence for intra-network competition.

Next, I test whether the effects of social ties on economic performance vary with the size of the network. As the size of the network increases, the number of job candidates becomes larger, and, hence, the competition becomes more intense. As a result, the promotion incentive for officials in a larger social group would be ***weaker***, and the economic performance would be worse. In light of this, I interact the social tie dummies with the size of the social group. The results in Table 7 show that the coefficients on the interaction terms are negative, which is in line with the expectation. In Table B8, I interact the social tie variables with the size of the social group for members who have the same job rank and find that the coefficients on the interaction terms are larger since the members in the social group are those who engage in direct competition with official i.[22]

Next, I investigate whether the rank of the corrupt officials with whom the individual has social ties matters for economic performance. If the story of intra-network competition is correct, only the corruption investigation of higher-ranked officials will initiate promotion competition, and boost economic performance. In the regressions, I replace the social tie dummy with two dummies, one that is equal to 1 if the investigated corrupt officials are higher-ranked, and the other that is equal to 1 if they are lower-ranked. Only the coefficients on the dummy variable with a higher rank should be positive, and their marginal effect should be larger than the dummy with a lower rank. Table 8 shows that this is just the case.

[22] As Fisman et al. (2020) and Jiang et al. (2022a) argue, politicians in the same social networks may be subject to more intense political competition because of the quota of promotion they face. Therefore, the size of the network measures the intensity of the network-based competition and promotion incentives.

Table 7
The effects of group size.

	(1)	(2)	(3)	(4)	(5)	(6)
	per capita GDP growth rates (%)					
1(having college tie to corrupt officials)	4.011***	4.163***	4.067***			
	(1.210)	(1.318)	(1.286)			
group size	−3.316	−0.399	−0.698	0.00573	0.00793	0.00738
	(3.266)	(4.580)	(4.578)	(0.00670)	(0.00939)	(0.00937)
1(having college tie to corrupt officials)* group size	−0.0925*	−0.112**	−0.123**			
	(0.0499)	(0.0569)	(0.0584)			
1(having workplace tie to corrupt officials)				5.251**	1.603	1.096
				(2.398)	(2.406)	(2.472)
1(having workplace tie to corrupt officials)* group size				−0.0165	−0.00257	−0.000901
				(0.0104)	(0.0124)	(0.0127)
College FE	Y	Y	Y	N	N	N
Workplace FE	N	N	N	Y	Y	Y
City FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Official Controls	N	Y	Y	N	Y	Y
City Controls	N	N	Y	N	N	Y
Observations	2,803	2,553	2,511	2,803	2,553	2,511
R-squared	0.328	0.333	0.380	0.296	0.300	0.335

Notes: The sample covers 935 officials, 287 cities, and 5 years from 2013 to 2017. Official controls include the age, a dummy for college and a dummy for graduate school, ethnicity dummies, and a gender dummy of the officials. City controls include the lag of log population and the lag of GDP share of secondary industry. * Significant at 10%, ** 5%, *** 1%. The standard errors are clustered at the city level.

Table 8
Heterogenous effects of job ranks.

	(1)	(2)	(3)	(4)	(5)	(6)
	per capita GDP growth rates (%)					
1(having college tie to investigated officials with higher rank)	2.188**	1.852*	1.828*			
	(1.031)	(1.072)	(1.101)			
1(having college tie to investigated officials with lower rank)	1.106	1.292	1.129			
	(1.030)	(1.108)	(1.153)			
1(having workplace tie to investigated officials with higher rank)				1.872	3.176*	2.879*
				(1.914)	(1.718)	(1.724)
1(having workplace tie to investigated officials with lower rank)				0.781	0.0830	0.241
				(1.683)	(1.743)	(1.761)
College FE	Y	Y	Y	N	N	N
Workplace FE	N	N	N	Y	Y	Y
City FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Official Controls	N	Y	Y	N	Y	Y
City Controls	N	N	Y	N	N	Y
Observations	2,803	2,553	2,511	2,803	2,553	2,511
R-squared	0.328	0.333	0.379	0.296	0.300	0.335

Notes: The sample covers 935 officials, 287 cities, and 5 years from 2013 to 2017. Official controls include the age, a dummy for college and a dummy for graduate school, ethnicity dummies, and a gender dummy of the officials. City controls include the lag of log population and the lag of GDP share of secondary industry. * Significant at 10%, ** 5%, *** 1%. The standard errors are clustered at the city level.

Then, I test whether the ranking of economic performance matters for the transmission of anti-corruption shocks. If the mechanism of intra-network competition is at play, then we should expect city officials who lag behind in the promotion competition to have ***stronger*** incentives to promote economic performance. To this end, I interact the connection dummies with the ranking of GDP growth in the same province.[23] The results, shown in Table A4 indicate that the city official behind in the promotion competition has better economic performance. Such results provide evidence that the mechanism of intra-network competition is at play.

Next, I examine whether the promotion incentives and the economic performance are affected by time or, more specifically, by the number of years till the upcoming reshuffling of government officials around National Two Sessions in 2017. As 2017 approaches, the promotion incentives become more salient, and, thus, city officials are more incentivized to boost economic growth. I interact the social tie variables with the number of years till 2017. Results in Table A5 show that the farther away 2017, the smaller the GDP growth rates; in 2017, the growth rates are approximately 3.0–8.0 larger than those in 2013. The results are, hence, consistent with the expectation. All of these results are consistent with Hypothesis 3.

[23] Per capita GDP growth ranking is defined as $Ranking = \frac{g_{ipt} - mean(g_{ipt})}{sd(g_{ipt})}$, where g_{ipt} is the growth rate of city i in province p and year t; $mean(g_{ipt})$ is the mean of all cities' growth rates in province p; and $sd(g_{ipt})$ is the standard deviation of all cities' growth rates in province p.

As the final test for the mechanism of intra-network competition, I examine whether the effects of social ties to corrupt peers on economic performance are mainly driven by those who work in specialized jobs. Officials who work in such jobs tend to have similar backgrounds and, thus, there will be stronger network "persistence", or more specifically, those jobs might be occupied by officials from the same social network. Such network persistence in ministries may stem from the requirement of specific skills for officials who work in these positions. The jobs in ministries (such as economic and financial, environmental, and judicial ministries) are those jobs that can meet such criteria. Therefore, I interact the social tie variables with a dummy variable indicating whether the connected corrupt peers work in a specialized ministry and estimate the baseline specification. The results reported in Table B7 suggest that network persistence matters: for college ties, the effects are stronger if the connected corrupt peers work in ministries, and it is less so for workplace ties. These results are consistent with the observation that officials in ministries often have similar educational backgrounds.

It is also possible that the removal of a corrupt official may serve as an "alert" to all of his social peers so that they have to work harder to keep their position. If this story is true, then we may expect that officials connected with high-ranking officials are less alerted because their connections can provide protection. I test this hypothesis by interacting the social tie dummies with another indicator variable of whether the official has birthplace, college, or workplace connections with any national-level officials. The results in Table A7 imply that officials connected with high-rank peers do not have a better performance after the removal of corrupt officials in their social networks. Thus, the story that the alert of removal of corrupt peers incentivizes city officials lacks empirical support. In addition, connected city leaders may have a stronger incentive to work harder to compensate for the loss of their political capital, namely their investigated social connections. To test this argument, I interact the total number of political connections, as a measure of political capital, with the dummy of connected anti-corruption investigations. According to Table A8, the coefficient of the interaction term is not negative and statistically significant at the same time, indicating that the loss of political capital is not a cause of the increase in economic performance.

There is another channel through which being socially tied to investigated officials may influence a city official's economic performance: the connected corrupt officials may bring attention from the anti-corruption agency, and, thus, the city official is more likely to be implicated (Shi, 2021b).[24] As a result, the city official has weaker promotion incentives because once he is promoted, he must go through a thorough inspection of personal property to rule out corruption. This implies that the anti-corruption crackdown on social ties and the city's own economic growth should be *negatively* correlated, which is inconsistent with the main results of this paper.

To further show that the fear of being implicated does not affect growth, I design two empirical tests. First, I interact the social tie variables with the proportion of officials who have been investigated for corruption in the same social group, and then estimate the baseline specification with the interaction term. The social group with a larger proportion of corrupt officials will attract more attention from the anti-corruption agency. The results in Table A9 indicate that the coefficients on the interaction terms are not statistically significant, implying that the fear of being implicated does not affect promotion incentives. Second, I interact the social tie variables with the intensity of the corruption investigation in the corresponding province, measured by the number of corruption cases normalized by the provincial population. In provinces with more intense corruption investigations, the probability of being implicated in corruption is also larger. In Table A10, the results show that the coefficients on the interaction terms are also insignificant, pointing to the argument that the attention from the anti-corruption agency does not matter for economic performance.

There might be another explanation for the positive correlation between having social ties with investigated officials and economic growth: those socially connected officials are more capable of developing the local economy. I use two dimensions to proxy their capacity: education and experience. On the one hand, Besley et al. (2011) find that leaders with higher-level education degrees have better records of economic growth. To test this hypothesis, I regress a dummy variable equal to 1 if the official has a graduate degree on the social tie variables (and controls). Results in Table B9 indicate that such a hypothesis is invalid: all of the coefficients on the social tie variables are statistically insignificant.

On the other hand, I explore whether officials connected to their corrupt peers are more experienced. I count the number of the variety of different work experiences before becoming a city official, as the same fashion in Shi et al. (2020), who find that the variety of experiences leaders' of is positively correlated with economic growth. I use the variety of experiences as the dependent variable and estimate the same specification as in the previous analysis. Results in Table B10 show that the coefficients on the social tie variables are all statistically insignificant, suggesting that being connected to investigated officials is orthogonal to the experiences of the city officials. In all, Table B9 and Table B10 indicate that city officials who are socially connected to corrupt peers are not more capable of developing the local economy.

5.4. The costs of intra-network competition

Next, I explore the welfare effects of intra-network competition induced by anti-corruption. Specifically, I investigate whether anti-corruption affects other policy outcomes, especially long-term ones, in addition to economic growth.

First, I test whether anti-corruption investigations on connected officials will cause the city leader to pay less attention on innovation, which is a fundamental cause of long-run growth (Grossman and Helpman, 1993), but does not have significant short-run impacts. I use the share of language involving encouraging innovation in the city government work reports as the dependent variable

[24] In Shi (2021b), I argue that the spillover effects of corruption investigations on connected politicians shape peer effects of corruption. In this paper, I further argue that this mechanism may be also at play in shaping politicians' economic performance.

Table 9
Effects on innovation policy.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Share of innovation in government work report								
1(having city tie to corrupt officials)	−0.00813***	−0.00813***	−0.00816***						
	(0.000220)	(0.000245)	(0.000248)						
1(having college tie to corrupt officials)				−0.0125***	−0.0125***	−0.0127***			
				(0.000559)	(0.000574)	(0.000603)			
1(having workplace tie to corrupt officials)							−0.0111***	−0.0108***	−0.0107***
							(0.00109)	(0.000974)	(0.000966)
Home City FE	Y	N	N	Y	N	N	Y	N	N
College FE	N	Y	N	N	Y	N	N	Y	N
Workplace FE	N	N	Y	N	N	Y	N	N	Y
Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Official Controls	Y	Y	Y	Y	Y	Y	Y	Y	Y
City Controls	Y	Y	Y	Y	Y	Y	Y	Y	Y
Observations	2,764	2,534	2,478	2,823	2,565	2,506	2,834	2,566	2,505
R-squared	0.798	0.803	0.806	0.629	0.663	0.666	0.525	0.548	0.552

Notes: The sample covers 935 officials, 287 cities, and 5 years from 2013 to 2017. Official controls include the age, ethnicity dummies, and a gender dummy of the officials. City controls include the lag of log population and the lag of GDP share of secondary industry. * Significant at 10%, ** 5%, *** 1%. The standard errors are clustered at the city level.

Table 10
The effects of corruption investigation on education expenditure.

	(1)	(2)	(3)	(4)	(5)	(6)
	share of education expenditure					
1(having college ties to investigated officials)	−0.00576**	−0.00653**	−0.00653**			
	(0.00290)	(0.00313)	(0.00313)			
1(having workplace ties to investigated officials)				−0.0312***	−0.0304***	−0.0303***
				(0.00278)	(0.00310)	(0.00314)
College FE	Y	Y	Y	N	N	N
Workplace FE	N	N	N	Y	Y	Y
City FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Official Controls	N	Y	Y	N	Y	Y
City Controls	N	N	Y	N	N	Y
Observations	2,830	2,518	2,518	2,830	2,576	2,518
R-squared	0.858	0.858	0.858	0.853	0.854	0.856

Notes: The sample covers 935 officials, 287 cities, and 5 years from 2013 to 2017. Official controls include the age, a dummy for college and a dummy for graduate school, ethnicity dummies, and a gender dummy of the officials. City controls include the lag of log population and the lag of GDP share of secondary industry. * Significant at 10%, ** 5%, *** 1%. The standard errors are clustered at the city level.

and rerun the baseline specification. The results are displayed in Table 9. Having connections with investigated officials significantly reduces the share of language concerning innovation in the government work report. Moreover, using granular firm-level data, I also find that firms tend to apply for fewer patents and spend less on R&D (Table B11). The effects are even more negative for politically connected firms,[25] indicating an improvement in government-business relationships and the business environment.

Second, I test whether intra-network competition affects the fiscal input in education, which matters for long-term economic development (Barro (2001)), but has little impact on short-term growth. To this end, I use the share of education expenditures in total fiscal expenditures as the dependent variable. Table 10 shows that, having college ties to investigated officials reduces the share of education expenditures by about 0.6%, which is about 3.4% of the sample mean. Having workplace ties to investigated officials has qualitatively similar effects. Finally, I also find that city leaders connected to investigated officials emphasize less on education in their government work report (Table B12).

Third, I explore the effects of having social ties with investigated officials on pollution, which is a byproduct of economic development. I use the log emission intensity, defined by the amount of pollutant emissions normalized by GDP, of three pollutants (wastewater, SO_2, and dust) as the dependent variable. In Table 11, the coefficients on the social tie variables are mostly positive and statistically significant. Such results indicate that the boost in GDP growth rates takes place at the expense of the environment. Table B14 implies that the pollution intensity is not linked with promotion, so the city's mayor and Party Secretary can harm the environment without damaging their promotion prospects. Finally, I also find that city leaders connected to investigated officials emphasize less on environmental protection in their government work report (Table B13).

[25] The definition of political connections is consistent with Shi et al. (2021), Shi (2021a). Using firm registration data, I define a firm to be politically connected (to the city leader) if the origin of its legal representative is the previous workplace or hometown of the city leader.

Table 11
The effects of corruption investigation on pollution.

	(1)	(2)	(3)	(4)	(5)	(6)
	log(wastewater EI)	log(SO2 EI)	log(dust EI)	log(wastewater EI)	log(SO2 EI)	log(dust EI)
1(having college ties to investigated officials)	0.100*	0.0968	0.258***			
	(0.0600)	(0.107)	(0.0991)			
1(having workplace ties to investigated officials)				0.0701	0.277*	0.484*
				(0.0631)	(0.155)	(0.267)
College FE	Y	Y	Y	N	N	N
Workplace FE	N	N	N	Y	Y	Y
City FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Official Controls	Y	Y	Y	Y	Y	Y
City Controls	Y	Y	Y	Y	Y	Y
Observations	2,377	2,390	2,307	2,377	2,390	2,307
R-squared	0.927	0.893	0.921	0.925	0.888	0.919

Notes: The sample covers 935 officials, 287 cities, and 5 years from 2013 to 2017. Official controls include the age, a dummy for college and a dummy for graduate school, ethnicity dummies, and a gender dummy of the officials. City controls include the lag of log population and the lag of GDP share of secondary industry. * Significant at 10%, ** 5%, *** 1%. The standard errors are clustered at the city level.

5.5. Robustness checks

In this subsection, I conduct three more robustness checks. First, I test whether the main results are robust with respect to different measures of social ties. I change the main independent variable to the number of social ties with officials having been investigated for corruption, and I estimate the baseline specification. Columns (1) through (3) in Table B15 show that an increase of one standard deviation in the number of college ties to investigated officials leads to an increase of the per capita GDP growth rate by 0.69 percentage points. Similarly, columns (4) through (6) in Table B15 show that an increase of one standard deviation in the number of workplace ties to investigated officials leads to an increase of the per capita GDP growth rate by 0.22 percentage points.

Next, I conduct a placebo test to check whether corrupt officials without social ties affect economic outcomes. I replace the independent variables in the previous regressions with the number of officials who have been investigated for corruption but do not have social ties, and then rerun the baseline specification. The results are shown in Table B16. In columns (1) through (3), the coefficients on the number of corrupt officials without college ties are very small and statistically insignificant. In columns (4) through (6), although the coefficients for corrupt officials without workplace ties are statistically significant, the magnitudes are very small; the marginal effects of an additional corrupt official increase per capita GDP growth by only 0.003%, about 0.03% of the sample mean, and is about 2% of the marginal effect of corrupt officials with workplace ties. Such results indicate that the promotion competition resulting from the corruption investigation takes place inside the social network.

Next, I exclude all investigated city mayors and city Party Secretaries from the sample, and redo the fixed-effects estimation. The results presented in Table B17 are still robust, indicating that our main results are not driven by the negative impacts of investigations compared to the uninvestigated connected city leaders.

Finally, we use an IV estimation to evaluate the effects of anti-corruption shocks on other outcomes, including fixed investments, fiscal expenditures, the share of educational expenditures in total fiscal expenditures, and SO2 emission intensity. The results reported in Table B18 suggest that the sign of the coefficient on the dummy variables of college ties and workplace ties is the same as that of the results of OLS estimation.

6. Conclusion

In this paper, I provide an empirical analysis of how anti-corruption affects economic growth in Chinese cities, focusing on a novel mechanism: intra-network competition. To be more specific, I examine how sharing college or workplace connections with officials who have been investigated for corruption affects a city's mayor or Party Secretary's economic performance and policy outcomes. Exploiting the anti-corruption crackdown as a quasi-natural experiment, and in specifications that include fixed effects to control for differences across colleges and workplaces, I find that college and workplace connections are associated with a 2.5-percentage-point increase in the per capita GDP growth rate, about 28% of the sample mean. The results still hold when exploiting an instrumental variable approach.

This effect exists only with higher-ranked officials who are investigated for corruption. The effect decreases as the size of the official's social group increases, and as the time is farther away from 2017. Such evidence supports the explanation that the corruption investigation of officials initiates promotion competition within the social network. In addition, I find that city officials promote economic growth by improving the business environment, attracting fixed investment, and increasing government expenditures, but at the expense of long-term goals such as innovation, education, and environmental protection. Therefore, social networks, interacting with the recent anti-corruption campaign, can have both a bright side (boosting economic growth) and a dark side (overlooking long-term issues such as innovation, education, and environmental protection), pointing to the dichotomous nature of social networks, political competition, and anti-corruption.

Anti-corruption is a crucial policy issue in many countries. This paper helps to provide an understanding of the political and economic consequences of a massive anti-corruption campaign, in the context of politicians' social networks and political competition for promotion. This paper focuses on China, but its main analysis could apply to many other countries. This paper also illustrates the mechanisms of political selection and promotion competition in the largest non-democracy in the world.

In this paper, I did not compare the benefits and costs associated with the unintended consequences of the transmission of the shocks of the anti-corruption campaign. I think that this will be a fruitful direction for future research.

CRediT authorship contribution statement

Xiangyu Shi: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ejpoleco.2024.102549.

References

Acemoglu, D., Carvalho, V.M., Ozdaglar, A., Tahbaz-Salehi, A., 2012. The network origins of aggregate fluctuations. Econometrica 1977–2016.

Acemoglu, D., Johnson, S., Robinson, J.A., 2001. The colonial origins of comparative development: An empirical investigation. Amer. Econ. Rev. 91 (5), 1369–1401.

Acemoglu, D., Johnson, S., Robinson, J.A., 2005. Institutions as a fundamental cause of long-run growth. In: Handbook of economic growth, vol. 1, Elsevier, pp. 385–472.

Acemoglu, D., Ozdaglar, A., Tahbaz-Salehi, A., 2015. Systemic risk and stability in financial networks. Amer. Econ. Rev. 105 (2), 564–608.

Allen, F., Gale, D., 2000. Financial contagion. J. Polit. Econ. 108 (1), 1–33.

Banerjee, A., Chandrasekhar, A.G., Duflo, E., Jackson, M.O., 2013. The diffusion of microfinance. Science 341 (6144).

Barro, R.J., 1990. Government spending in a simple model of endogenous growth. J. Polit. Econ. 98 (5 pt 2).

Barro, R.J., 2001. Education and economic growth. Contrib. Hum. Soc. Cap. Sustain. Econ. Growth Well-Being 14–41.

Benoit, K., Giannetti, D., 2008. Intra-party politics and coalition governments: Concluding remarks. In: Intra-Party Politics and Coalition Governments. Routledge, pp. 245–252.

Besley, T., Montalvo, J.G., Reynal-Querol, M., 2011. Do educated leaders matter? Econ. J. 121 (554), F205–227.

Bosker, M., Brakman, S., Garretsen, H., De Jong, H., Schramm, M., 2008. Ports, plagues and politics: explaining Italian city growth 1300–1861. Eur. Rev. Econ. History 12 (1), 97–131.

Boucek, F., 2009. Rethinking factionalism: typologies, intra-party dynamics and three faces of factionalism. Party Politics 15 (4), 455–485.

Brumfiel, E.M., Fox, J.W., 2003. Factional competition and political development in the New World. Cambridge University Press.

Card, D., 1999. The causal effect of education on earnings. In: Handbook of labor economics, vol. 3, Elsevier, pp. 1801–1863.

Card, D., 2001. Estimating the return to schooling: Progress on some persistent econometric problems. Econometrica 69 (5), 1127–1160.

Chen, S.X., 2022. Vulnerability to tax enforcement and spillovers of corruption: cross-industry evidence from China. J. Law Econ. Organiz..

Chen, T., Hong, J.Y., 2016. The Enemies Within: Loyalty, Faction and Elite Competition under Authoritarianism. Technical report, Working paper.

Chen, Z., Jin, X., Xu, X., 2021. Is a corruption crackdown really good for the economy? Firm-level evidence from China. J. Law Econ. Organiz. 37 (2), 314–357.

Da Mata, D., Deichmann, U., Henderson, J.V., Lall, S.V., Wang, H.G., 2007. Determinants of city growth in Brazil. J. Urban Econ. 62 (2), 252–272.

d'Agostino, G., Dunne, J.P., Pieroni, L., 2016. Government spending, corruption and economic growth. World Dev. 84, 190–205.

De Long, J.B., Shleifer, A., 1993. Princes and merchants: European city growth before the industrial revolution. J. Law Econ. Organiz. 36 (2), 671–702.

Ding, H., Fang, H., Lin, S., Shi, K., 2020. Equilibrium consequences of corruption on firms: Evidence from China's anti-corruption campaign. Technical report, National Bureau of Economic Research.

Elliott, M., Golub, B., Jackson, M.O., 2014. Financial networks and contagion. Amer. Econ. Rev. 104 (10), 3115–3153.

Fisman, R., Shi, J., Wang, Y., Wu, W., 2020. Social ties and the selection of China's political elite. Amer. Econ. Rev. 110 (6), 1752–1781.

Fisman, R., Wang, Y., 2015. Corruption in Chinese privatizations. J. Law Econ. Organiz. 31 (1), 1–29.

Francois, P., Trebbi, F., Xiao, K., 2016. Factions in nondemocracies: Theory and evidence from the chinese communist party. Technical report, National Bureau of Economic Research.

Grossman, G.M., Helpman, E., 1993. Innovation and growth in the global economy. MIT Press.

Gründler, K., Potrafke, N., 2019. Corruption and economic growth: New empirical evidence. Eur. J. Political Econ. 60, 101810.

Henderson, J.V., Wang, H.G., 2007. Urbanization and city growth: The role of institutions. Reg. Sci. Urban Econ. 37 (3), 283–313.

Hong, J.Y., Yang, L.Y., 2018. Factional competition and power sharing under authoritarianism.

Huang, J., 2006. Factionalism in Chinese communist politics. Cambridge University Press.

Jia, R., Kudamatsu, M., Seim, D., 2015. Political selection in China: The complementary roles of connections and performance. J. Eur. Econom. Assoc. 13 (4), 631–668.

Jiang, J., 2018. Making bureaucracy work: Patronage networks, performance incentives, and economic development in China. Am. J. Political Sci. 62 (4), 982–999.

Jiang, Z., Shi, X., Wu, X., You, J., 2022a. Fly own colors: Local leaders' hometown connections and regional policy divergence. Available at SSRN.

Jiang, Z., Shi, X., Wu, X., You, J., 2022b. The social network origin of state capacity. Available at SSRN.

Jiang, J., Zhang, M., 2020. Friends with benefits: Patronage networks and distributive politics in China. J. Public Econ. 184, 104143.
Jones, B.F., Olken, B.A., 2005. Do leaders matter? National leadership and growth since world war II. Q. J. Econ. 120 (3), 835–864.
Keller, F.B., 2016. Moving beyond factions: using social network analysis to uncover patronage networks among Chinese elites. J. East Asian Stud. 16 (1), 17–41.
Khan, A.Q., Khwaja, A.I., Olken, B.A., 2019. Making moves matter: Experimental evidence on incentivizing bureaucrats through performance-based postings. Amer. Econ. Rev. 109 (1), 237–270.
Lambsdorff, J.G., 2006. Causes and consequences of corruption: What do we know from a cross-section of countries. In: International handbook on the economics of corruption, vol. 1, Edward Elgar Cheltenham, pp. 3–51.
Lan, X., Li, W., 2018. Swiss watch cycles: Evidence of corruption during leadership transition in China. J. Comp. Econ. 46 (4), 1234–1252.
Levine, R., Renelt, D., 1992. A sensitivity analysis of cross-country growth regressions. Am. Econ. Rev. 942–963.
Li, C., 2014. Xi Jinping's inner circle (part 1: the Shaanxi gang).
Li, H., Zhou, L.-A., 2005. Political turnover and economic performance: the incentive role of personnel control in China. J. Public Econ. 89 (9–10), 1743–1762.
Liu, Y., Shi, X., 2023. The dichotomy of social networks: Politicians' hometown ties and intercity investment in China. Available at SSRN 4390251.
Lorentzen, P.L., Lu, X., 2018. Personal ties, meritocracy, and China's anti-corruption campaign. Meritocracy, and China's Anti-Corruption Campaign (November 21, 2018).
Manion, M., 1996. Corruption by design: Bribery in Chinese enterprise licensing. J. Law Econ. Organiz. 12 (1), 167–195.
Mo, P.H., 2001. Corruption and economic growth. J. Compar. Econ. 29 (1), 66–79.
Qian, N., Wen, J., 2015. The impact of Xi Jinping's anti-corruption campaign on luxury imports in China. Prelim. Draft Yale Univ..
Qu, G., Sylwester, K., Wang, F., 2018. Anticorruption and growth: Evidence from China. Eur. J. Political Econ. 55, 373–390.
Reiter, H.L., 2004. Factional persistence within parties in the United States. Party Politics 10 (3), 251–271.
Shi, X., 2021a. Hometown favoritism and intercity investment networks in china. Available at SSRN 3745064.
Shi, X., 2021b. Peer effects, social ties, and corruption: Evidence from china. Social Ties, and Corruption: Evidence from China (March 11, 2021).
Shi, X., 2021c. Reevaluating the productivity effects of the environmental regulation through the lens of firm dynamics: the case of china's eleventh five-year plan. Available at SSRN, 3943772.
Shi, X., Wang, C., 2022. Carbon emission regulation, input-output networks, and firm dynamics: The case of low-carbon zone pilot in china. Available at SSRN 4004340.
Shi, X., Xi, T., Yao, Y., 2020. Better than on-the-job training: National executive's political experience and economic performance.
Shi, X., Xi, T., Zhang, X., Zhang, Y., 2021. "Moving umbrella": Bureaucratic transfers and the comovement of interregional investments in China. J. Dev. Econ. 153, 102717.
Shi, X., Zhibo, T., Zhang, X., 2022. Air pollution and firm dynamics: the case of chinese manufacturing. Available at SSRN 4002894.
Shih, V., 2016. Efforts at exterminating factionalism under Xi Jinping: Will Xi Jinping dominate Chinese politics after the 19th party congress? China's Core Execut. 18.
Shih, V., Adolph, C., Liu, M., 2012. Getting ahead in the communist party: explaining the advancement of central committee members in China. Am. Political Sci. Rev. 106 (1), 166–187.
Shu, Y., Cai, J., 2017. "Alcohol bans": Can they reveal the effect of xi jinping's anti-corruption campaign? Eur. J. Political Econ. 50, 37–51.
Wang, E., 2019. Frightened mandarins: The adverse effects of fighting corruption on local bureaucracy. Available at SSRN 3314508.
Xu, C., 2011. The fundamental institutions of China's reforms and development. J. Econ. Lit. 49 (4), 1076–1151.
Xu, G., Yano, G., 2017. How does anti-corruption affect corporate innovation? Evidence from recent anti-corruption efforts in China. J. Comp. Econ. 45 (3), 498–519.
Yao, Y., Zhang, M., 2015. Subnational leaders and economic growth: evidence from Chinese cities. J. Econ. Growth 20 (4), 405–436.
Zhuang, M., 2019. Intergovernmental conflict and censorship: Evidence from China's anti-corruption campaign. Available at SSRN 3267445.